

IPC-2221 Overview (Engineering Summary)

Introduction

IPC-2221 is a widely recognized industry standard that provides general guidelines for printed circuit board (PCB) design. It covers topics such as conductor spacing, trace width, creepage distances, manufacturing tolerances, component placement, and documentation.

This document is a project-specific engineering summary inspired by commonly accepted PCB design practices.

PCB Design Objectives

The primary design objectives for this project were:

- Compact PCB layout
 - Reliable power delivery
 - Easy manufacturability
 - Simple assembly
 - Low production cost
 - Easy debugging
 - Electrical safety
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Trace Width Selection

Power traces carrying battery current were designed wider than signal traces.

Net Type Width

Power 0.80 mm

Signal 0.25 mm

Wider traces reduce voltage drop and improve thermal performance.

Copper Clearance

Adequate clearance was maintained between all electrical nets to satisfy KiCad Design Rule Check (DRC) requirements and reduce the risk of short circuits during fabrication.

Ground Plane

A continuous ground copper pour was added on the front copper layer.

Benefits include:

- Reduced ground impedance

- Improved return current path
 - Reduced electrical noise
 - Simplified routing
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Thermal Relief

Ground pads connected to the copper pour use thermal relief spokes.

Advantages:

- Easier soldering
 - Better heat control
 - Improved manufacturability
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Component Placement

Components were grouped according to their function.

Power path:

Battery Holder

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PTC Fuse

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Schottky Diode

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Output Connector

The voltage sensing circuit was placed close together to minimize analog noise.

Silkscreen

Reference designators and component values were positioned to remain readable after assembly.

Silkscreen was kept clear of pads and vias.

Design Rule Check

Before manufacturing, the PCB successfully passed:

- Electrical Rule Check (ERC)
- Design Rule Check (DRC)

Result:

Errors: **0**

Warnings: **0**

Manufacturing Outputs

The following files were generated:

- Gerber Files
- Excellon Drill Files
- Bill of Materials
- PCB Images
- Schematic PDF

Summary

This PCB follows commonly accepted PCB design practices with emphasis on manufacturability, readability, reliability, and ease of assembly.